Huffman Coding Algorithm

Abdullah All Mamun Siam

Lecturer, Dhaka International University(DIU)

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Introduction to Greedy Algorithm

Huffman coding is a greedy approach based lossless data compression algorithm. It uses variable-length encoding to compress the data. The main idea in Huffman coding is to assign each character with a variable length code. The length of each character is decided on the basis of its occurrence frequency. The character that occurred most time would have the smallest code and the character that occurred least in the data would have the largest code.

Huffman Coding Algorithm

Huffman Coding step-by-step:

Step-1: Calculate the frequency of each string.

Step-2: Sort all the characters on the basis of their frequency in ascending order.

Step-3: Mark each unique character as a leaf node.

Step-4: Create a new internal node.

Step-5: The frequency of the new internal node as the sum of the single leaf node

Step-6: Mark the first node as this left child and another node as the right child of the recently created node.

Step-7: Repeat all the steps from step-2 to step-6.

Example of Huffman Coding Algorithm

Apply the Huffman coding algorithm to this string

"BBBAAADDDDDDDDDCCCAAAABBBBBBBEEEEEDDDDAAACCCBBBEEE"

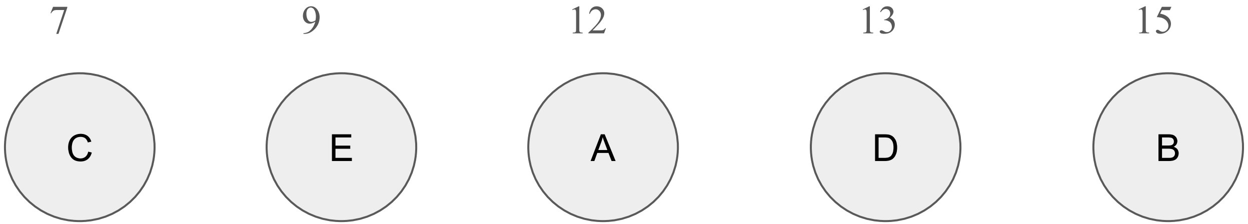
Solution:

Characters	Frequencies
A	12
B	15
C	7
D	13
E	9

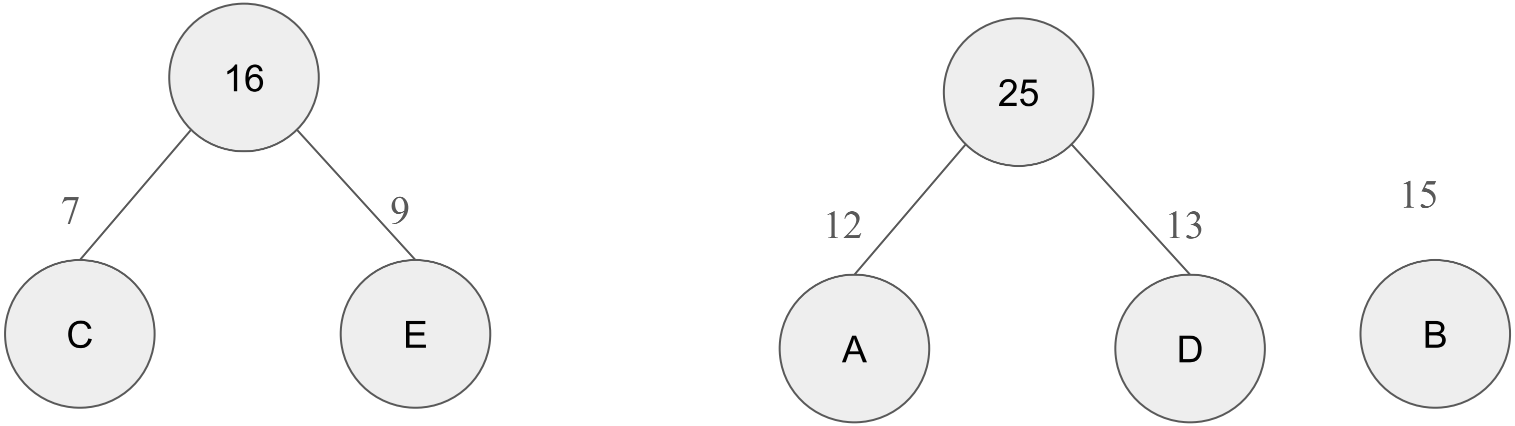
Example of Huffman Coding Algorithm

Characters	Frequencies
C	7
E	9
A	12
D	13
B	15

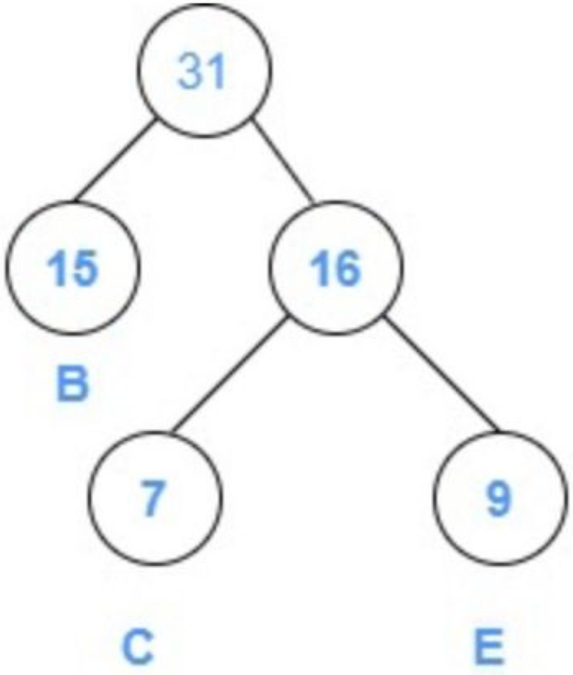
Example of Huffman Coding Algorithm

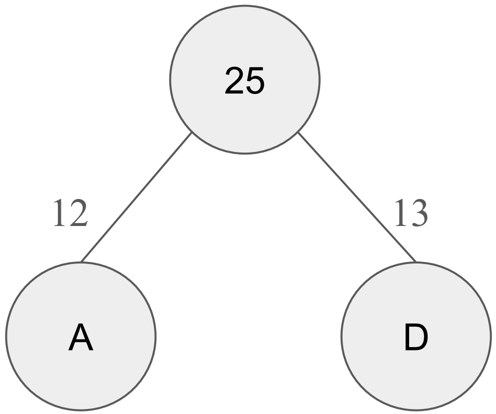


Example of Huffman Coding Algorithm

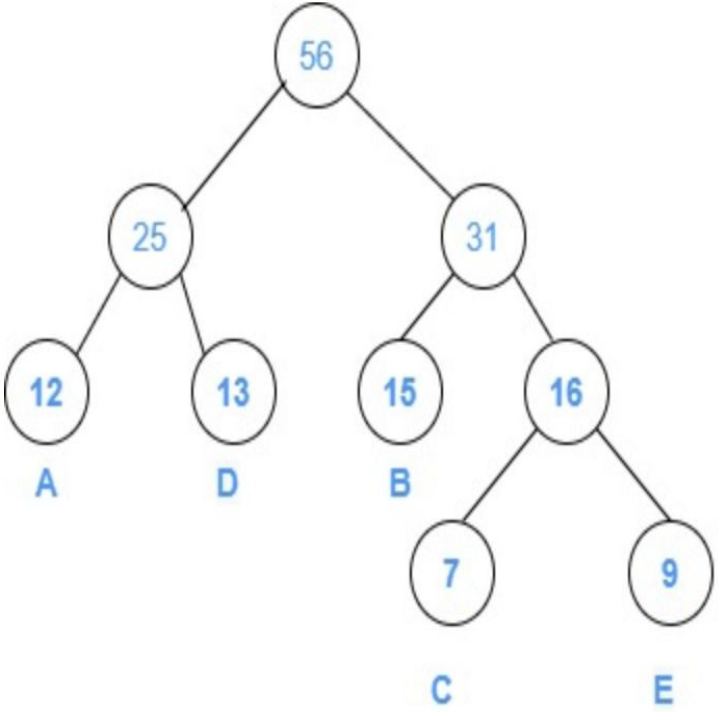


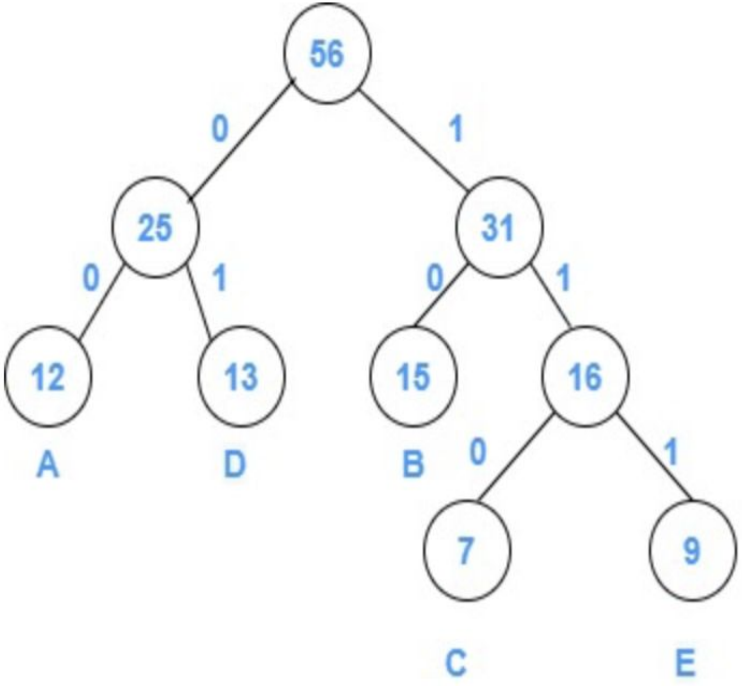
Example of Huffman Coding Algorithm





Example of Huffman Coding Algorithm





Example of Huffman Coding Algorithm

Total number of bits in Huffman encoded message$=\sum$(frequency x code length)

$= \{(12 \times 2) + (13 \times 2) + (15 \times 2) + (7 \times 3) + (9 \times 3)\}$

$=24+26+30+21+27$

$=128$

Average code length per character $= \sum$(frequency x code length)/ $\sum$ frequency

$= \{(12 \times 2) + (13 \times 2) + (15 \times 2) + (7 \times 3) + (9 \times 3)\} / (12 + 13 + 15 + 7 + 9)$

$= (24 + 26 + 30 + 21 + 27)/56$

$= 128/56$

$= 2.28$

Average code length per character = 2.28

Application of Huffman Coding

1.Huffman coding is used in traditional compression techniques like GZIP etc.

2.For tax and fax transmissions